

Guillem Masdemont Serra

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EDUCATION

Erasmus Mundus Joint Master's in Artificial Intelligence (EMAI)

Sept 2025 - Present

Sapienza Università di Roma / UL / UPF (Current grade: 9.6/10.00)

Italy/Slovenia/Spain

- Research-oriented Master's program focused on Artificial Intelligence delivered by a consortium of European universities. Acceptance rate $< 2\%$.

Bachelor's degree in Mathematics and Physics

Sept 2020 - Sept 2025

Autonomous University of Barcelona (Grade: 8.9/10.00)

Bellaterra, Spain

- Thesis in Mathematics: Application of a Spectral Method to Two-Dimensional Incompressible Flows in a Rectangular Domain. Supervisor: Prof. Jezabel Curbelo (UPC). Awarded Highest Honors.
- Thesis in Physics: Far field heat transfer and persistent heat currents arising in a cylindrical non-reciprocal configuration. Supervisor: Prof. Georgia Papadakis (ICFO). Awarded 9.4/10.

ACADEMIC EXPERIENCE

ICFO - La Pedrera Summer Research Fellow

Castelldefels, Spain

The Institute of Photonic Sciences (ICFO), La Pedrera Foundation

July 2024 - September 2024

- Research group: Thermal Photonics (Prof. Dr. Georgia Papadakis).
Investigated the breakdown of Kirchhoff's law in non-reciprocal systems to achieve directional heat flow control. Implemented a computational framework to optimize heat rectification and achieved theoretical benchmarks for perfect rectification and thermal circulation.

BIMR - Barcelona Introduction to Mathematical Research

Barcelona, Spain

Center of Mathematical Research, BGSMath

July 2023 - September 2023

- Research Group: Dynamical Systems group at University of Barcelona.
Investigated the iteration of transcendental entire functions under the supervision of Prof. Núria Fagella and Prof. Xavier Jarque. Delivered a technical presentation on Eremenko's 1989 work, which consisted on detailing the topological properties of the escaping set for transcendental functions.

PUBLICATIONS

Far-field heat transfer and monochromatic thermal currents in a cylindrical nonreciprocal cavity

G. Masdemont, J. Legendre, G. T. Papadakis. *Accepted in Physical Review Research*

- Developed a numerical algorithm to simulate far-field radiative heat transfer between non-planar, non-reciprocal bodies in cylindrical geometries and numerically showed that non-reciprocity can be tuned to achieve directional heat control. This establishes theoretical benchmarks for perfect thermal rectification.
- Article: <https://doi.org/10.1103/5bbx-qx7v> | Code: GitHub/Nonreciprocal-Cavity-Paper

AWARDS AND SCHOLARSHIPS

Erasmus Mundus Full Tuition Fee Waiver Scholarship

Sept 2025

Awarded by: European Union and the EMAI consortium

- Awarded a full exemption from tuition fees for the entire duration of EMAI master's program.

Distinction Honors in Baccalaureate Studies

June 2020

Awarded by: Lacetània High School

Grade: 9.91 / 10

- Top academic performance for the 2018-2020 graduating class.
- Pau/University admission exam: 13.51/14.00.

Cangur Mathematics Competition

March 2019

Awarded by: Catalan Society of Mathematics (SCM) and UPC

Position: 2nd place

- Recognized for high-level problem solving in the regional mathematics competition hosted at the Polytechnic University of Catalonia in Manresa.

OUTSIDE ACADEMIA

Violist

- Currently performing as a violist in the Camerata Bacasis Orchestra, under a semiprofessional musician contract. Our website is [here](#).

Student life

- Delegate of the Mathematics and Physics double degree, 2020-2025 promotion.
- Passionate about hackathons and math/physics competition. Awarded the ECIU Creathon Grant (May 2024) to represent UAB in Linköping (Sweden) for developing European smart city digital services.
- GLEE Mission: Collaborated on the Great Lunar Expedition for Everyone (Colorado Space Grant Consortium) to develop and test LunaSats for lunar surface experiments.

Personal Interests

- I enjoy hiking and cooking. More about myself can be found at guillemmasdemont.github.io